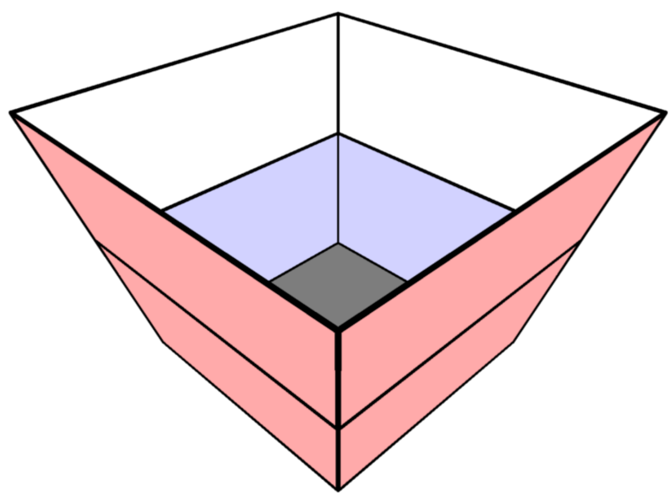
2025 AMC 10B Problem 19 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10B Problem 19. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10B Problem 19

Problem:

A container has a 1×1 square bottom, a 3×3 open square top, and four congruent trapezoidal sides, as shown. Starting when the container is empty, a hose that runs water at a constant rate takes 35 minutes to fill the container up to the midline of the trapezoids. How many more minutes will it take to fill the remainder of the container?



Answer Choices:

A. 70

B. 85

C. 90

D. 95

E. 105

Join the Discussion
Stuck on this problem or want to share your approach?
Continue the conversation and see what others are thinking: View Forum Thread

Solution:

The midlines of each trapezoid form a square with side length 2. Extend the legs of each trapezoidal face to meet at a point O, the vertex of the formed square pyramid. Consider A, B, and C to be 3 square pyramids with vertex O and bases as the 1×1, 2×2, and 3×3 squares respectively.

Suppose the volume of A is x. Since A, B, and C are similar solids, we know the ratio of their volumes is their scale factor cubed. Therefore we may say that $B = 8x$ and $C = 27x$.

The volume of water filled so far is $B - A = 7x$ and the volume of water left to be filled is $C - B = 19x$. Since water fills proportionally to volume, we may find the number of minutes as $\frac{19x}{7x} \cdot 35 = \boxed{\textbf{(D) 95}}$ minutes.